

Curriculum Vitae

Marco Bilucaglia, Ph.D., C.Eng.

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Summary

Marco Bilucaglia earned the B.Sc. (*Laurea*) in Electronics Engineering from Politecnico di Milano (Milan, Italy) in 2014, the M.Sc. (*Laurea Magistrale*) in Electronics Engineering (Bioelectronics curriculum) from Alma Mater Studiorum – Università di Bologna (Bologna, Italy) in 2018 and the Ph.D. in Communication, Markets and Society from Università IULM (Milan, Italy) in 2026. He completed the Italian State Examination (License to practise) for Professional Engineer (*Esame di Stato*) at Politecnico di Milano in 2020.

He has been a Ph.D. Fellow (2022–2025) and a Research Fellow (2019–2022) at Department of Business, Law, Economics and Consumer Behaviour “Carlo A. Ricciardi” of Università IULM (Milan, Italy). He has worked as full-time Bioengineer at Behavior and Brain Lab IULM – Neuromarketing Research Center of Università IULM since 2018.

He is co-author of several peer-reviewed papers published in journals and international conference proceedings. His research interests include Biomedical Signal Processing and Machine Learning with applications to Consumer Neuroscience and Human Performance.

He currently serves as referee for several journals, including *Frontiers in Psychology*, *PLOS ONE*, *Cognitive Neurodynamics* and *Scientific Reports*. He has been a Review Editor (Community Reviewer) for *Frontiers in Human Neuroscience* since 2026 and guest Associate Editor for Research Topics hosted by *Frontiers in Human Neuroscience*, *Frontiers in Neuroscience* and *Frontiers in Psychology* since 2023.

He has been member of the IEEE since 2008, and of the IEEE-EMBS since 2018. He has been a C.Eng. in Information Engineering of the Engineers Council of Milan (Italy) since 2020.

1 Anagraphic

- **Name & Surname:** Marco Bilucaglia
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3 Education

- **2026:** Ph.D. in Communication, Markets & Society, Università IULM (Milan, Italy)
Thesis: Affect Detection in Consumer Neuroscience: a Multimodal Machine Learning Model based on bioelectrical and biometric signals
Supervisor: Prof. V. Russo, Università IULM (Milan, Italy)
Co-supervisor: Prof. L. T. Mainardi, Politecnico di Milano (Milan, Italy)
- **2018:** M.Sc. (*Laurea Magistrale*) in Electronics Engineering (Bioelectronics curriculum), Alma Mater Studiorum – Università di Bologna (Bologna, Italy)
Thesis: Collection of stimulus–response pairs for Brain-Computer Interface applications (*Raccolta di coppie stimolo-risposta per applicazioni di Brain-Computer Interface*)
Supervisor: Prof. R. Rovatti, Alma Mater Studiorum – Università di Bologna
Co-supervisor: Prof. M. Mangia, Alma Mater Studiorum – Università di Bologna
- **2014:** B.Sc. (*Laurea*) in Electronics Engineering, Politecnico di Milano (Milan, Italy)

4 Professional Qualifications

- **2020:** Italian State Examination (License to Practice) for Professional Engineer (*Abilitazione all'Esercizio della Professione di Ingegnere*), Politecnico di Milano (Milan, Italy)

5 Professional Affiliations

- **2020–ongoing:** C.Eng. (*Ing.*) in Information Engineering, Engineers Council of Milan (Italy)
- **2018–ongoing:** Member, IEEE-EMBS (New York, NY, USA)
- **2014–ongoing:** Member, IEEE (New York, NY, USA)

6 Experience

Research Experience

All positions were obtained through competitive public selection procedures.

- **2022–2025:** PhD Fellow (Supervisor: Prof. V. Russo), Department of Business, Law, Economics and Consumer Behaviour “Carlo A. Ricciardi”, Università IULM (Milan, Italy)

Project: Machine Learning in the Study of Consumer Behaviour: An Affect Detection Model for Neuromarketing Applications (*Il Machine-Learning nello studio dei comportamenti di consumo: un modello di Affect Detection per applicazioni di Neuromarketing*)

- **2019–2022:** Research Fellow (Supervisor: Prof. V. Russo), Department of Business, Law, Economics and Consumer Behaviour “Carlo A. Ricciardi”, Università IULM (Milan, Italy)

Project: Study of psychophysiological, bioelectrical, and electroencephalographic signals for neuromarketing applications in the dairy sector, and the use of machine learning for the classification of emotional states and their integration with data from traditional market research – AGER Project (*Studio dei segnali psicofisiologici, bioelettrici ed elettroencefalografici per le applicazioni di neuromarketing nel settore caseario e di “Machine Learning” per la classificazione degli stati emozionali e l’integrazione con dati di ricerche di mercato tradizionali – Progetto AGER*)

- **2018:** Research Assistant (Supervisor: Dr. P. Tressoldi), Department of General Psychology, University of Padua (Padua, Italy)

Project: EEG signal analysis using machine-learning algorithms (*Analisi di dati EEG secondo algoritmi di Machine Learning*)

Work Experience

- **2018–ongoing:** Bioengineer, Behavior and Brain Lab IULM – Neuromarketing Research Center, Università IULM (Milan, Italy)

Teaching Experience

Master’s degrees

- **2022–ongoing:** Teaching Assistant (Instructor: Prof. V. Russo), “Laboratory of Cognition, Neuroscience and Machine Learning” [20 hours], M.Sc. (*Laurea Magistrale*) in Artificial Intelligence for Business and Society, Università IULM (Milan, Italy)
- **2022–ongoing:** Teaching Assistant (Instructor: Dr. A. Fici), “Neuroscience for Communication Lab” [10 hours], M.Sc. (*Laurea Magistrale*) in Strategic Communication, Università IULM (Milan, Italy)

Executive education

- **2022–ongoing:** Instructor, “Introduction to Big Data and Machine Learning” [4 hours], “EEG metrics for emotion, engagement and workload measurement” [4 hours], Executive Program (*Master Executive*) in Neuromarketing and Consumer Neuroscience, Università IULM (Milan, Italy)

7 Academic Services

Editorial roles

- **2026 – present:** Review Editor (Community Reviewer), Frontiers in Human Neuroscience

Guest Editor for Research Topics / Special Issues

- **2026–present:** “Machine-Learning/Deep-Learning methods in Neuromarketing and Consumer Neuroscience – Volume III”, Frontiers in Human Neuroscience
- **2024–2026:** “Machine-Learning/Deep-Learning methods in Neuromarketing and Consumer Neuroscience – Volume II”, Frontiers in Human Neuroscience
- **2023–2024:** “Machine-Learning/Deep-Learning methods in Neuromarketing and Consumer Neuroscience”, Frontiers in Neuroscience, Frontiers in Human Neuroscience and Frontiers in Psychology (co-hosting)

Ad hoc Reviewer [manuscripts, reviews]

Frontiers in Psychology [8, 8], Frontiers in Public Health [7, 7], Cognitive Neurodynamics [6, 17], Scientific Reports [6, 16], HardwareX [4, 9], PLOS ONE [4, 9], Frontiers in Psychiatry [3, 3], Frontiers in Human Neuroscience [2, 2], BMC Research Notes [1, 2], Biocybernetics and Biomedical Engineering [1, 2], Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC) [1, 1], Cognitive Computation [1, 1], Frontiers in Child and Adolescent Psychiatry [1, 1], Frontiers in Physiology [1, 1], Scientific Data [1, 1]

Ad hoc Handling Editor [manuscripts]

Frontiers in Human Neuroscience [5], Frontiers in Medicine [2], Frontiers in Public Health [1], Frontiers in Social Psychology [1]

Conference Services

- **2025–2026:** Event Committee Member, “1st International Online Conference on Behavioral Sciences (IOCBS)”, Behavioral Sciences

8 Research Interests

Biomedical Signal Processing

This research line focused on the development of new algorithms for processing electroencephalographic (EEG) signals, as well as the evaluation of existing methods.

In [J25] we designed a novel method to detect the EEG Event-Related Potential (ERP) response, called “GW6”. This method is based on Pearson’s product-moment correlation computed within a sliding window. It produces a signal with the same temporal resolution as the classical averaged ERP, but shows only positive peaks corresponding to increases in correlation across the EEG channels. We showed that GW6 is able to recover hidden ERP components that are usually suppressed by classical averaging due to interstimulus phase jitter in individual trial responses.

In [C5] we investigated the influence of different computational methods on the computation of frontal Alpha Asymmetry (FAA) for estimating Emotional Valence. The effects of the number of channels, power computation method (filtering versus Short-Time Fourier Transform – STFT), and normalisation were assessed on the DEAP dataset by computing, for each combination, the average correlation between FAA and self-reported valence. Unlike the power computation method and normalisation, the number of channels showed no significant effect. The maximum correlation between FAA and valence was obtained using the STFT approach combined with normalisation.

Machine Learning

This research line focused on the development of Machine Learning (ML) algorithms for the classification of bioelectrical signals, with applications to affect detection and non-local brain-to-brain communication.

In [J23] we reanalysed two previously collected datasets to classify the EEG activity of a group of isolated participants (receivers) while another group of participants (senders) was exposed to audio and video stimulation according to the P300 and SSVEP paradigms. We extracted temporal and spectral EEG features and trained a Linear Discriminant Analysis (LDA) classifier. For robustness, cross-validated LDA accuracies were compared with those empirically produced by a random classifier. We showed that LDA was able to discriminate stimulation from non-stimulation trials, although the associated accuracy was weak but still significantly greater than chance.

In [J17] we reanalysed a previously collected dataset to classify the arousal level of visual and auditory stimuli from EEG activity during the pre-stimulus interval. Linear Discriminant Analysis (LDA), k-Nearest Neighbours (kNN), and Support Vector Machine (SVM) classifiers, implemented in static and dynamic forms, were trained using temporal and spectral features. For robustness, all obtained accuracies were compared with those empirically produced by static and dynamic random classifiers. Among the static and dynamic classifiers, the dynamic SVM achieved the highest accuracy.

In [C8] we designed a multimodal (EEG, SC and HR) affect-detection system based on low-cost wearable sensors. A 52-dimensional feature vector was created from EEG, SC and HR data recorded during exposure to 24 emotional pictures taken from the IAPS dataset. A 1-NN classifier was trained to predict subjective levels of valence (low vs. high) and arousal (low vs. high), rated using the SAM. Overall accuracy was higher for arousal than for valence. We showed that arousal classification performed best when all modalities were considered, whereas valence classification did not improve when SC and HR features were added to EEG.

In [J9] we presented I DARE, the IULM Dataset of Affective Responses, which includes the physiological responses of 63 participants exposed to 32 emotional pictures. The stimuli were selected

from the IAPS and OASIS datasets through semi-automatic procedures based on their geometrical properties in the arousal-valence plane and the ratings of nine independent judges. The modalities include EEG, SC, PPG, EMG and eye-tracking signals preprocessed to reduce external noise and artefacts. The dataset, which is publicly available (https://figshare.com/projects/I_DARE/186558), goes beyond the state of the art in terms of dimensions, stimulus type, and minimum detectable effect size.

In [J3] we trained multiple least-squares Boosted Trees (LSBoost) regressors using EEG data to predict the sensory attributes (acid, astringent, bitter, sweet) of coffee samples and basic water solutions. We extracted spectral and temporal features during tasting tasks and optimised LSBoost’s hyperparameters using a Bayesian approach. Overall, the models achieved high predictive accuracy and robust performance relative to benchmark regressors. Feature-importance analysis identified alpha and beta spectral power, together with Hjorth’s activity and complexity within central and frontal brain regions, as the most predictive features.

In [J4] we presented the Research Topic “Machine-Learning/Deep Learning Methods in Neuro-marketing and Consumer Neuroscience” and summarised the collected contributions (6 experimental studies, 1 dataset paper and 1 meta-analysis). The Research Topic ran from 28 December 2022 to 14 January 2024 and was co-hosted by *Frontiers in Human Neuroscience*, *Frontiers in Psychology*, *Frontiers in Neuroscience* and *Frontiers in Computational Neuroscience*.

Human Performance

This research line focused on the analysis of EEG, SC and HR data for the assessment of human performance, with applications to human factors, ergonomics and personal growth (i.e., coaching).

In [C9], [A32], we tested the effects of wearing blue lenses on increasing wakefulness after a relaxing condition. After inducing a calming state using Mozart’s classical music, we administered a reactivity test (PVT) while EEG, SC and pupil-diameter data were collected. We compared the recordings of the experimental sample (wearing the lenses during the PVT) with those of the control sample (not wearing the lenses during the PVT) using a between-group design. We confirmed an awakening effect of the blue lenses, reflected in a decrease in GFP in the theta band and an increase in SCR, as well as a strong negative correlation between pupil diameter and the experimental condition.

In [C6], [A32] we tested the effects of wearing yellow lenses on concentration during a simulated driving task (HPT). Using a between-group design, we compared several EEG-, SC- and gaze-related metrics between the experimental sample (wearing the lenses during the HPT) and the control sample (not wearing the lenses during the HPT). In the experimental sample, we confirmed a focusing effect, reflected in increases in first fixation duration and BAR, as well as decreases in distraction, workload, and the number of gaze revisits.

In [A19], [A31], [C7] we recorded synchronised EEG and SC data from both the coach and the coachee during 36 individual coaching sessions conducted according to two different coaching methods: core coaching and system constellations. We investigated the temporal evolution of bonding and mutual engagement using a similarity metric based on the DTW distance between signals (namely, S-TVM). We found significant differences between session phases for EEG-related S-TVM measures (BAR, BATR, and AWI), with maximum values (defined as “tuning”) occurring in the same phase but differing between the two experiments. In [J13] we reanalysed data related to one of the above-mentioned coaching sessions (i.e., the Core Coaching approach) and reinterpreted the results according to the work-transition framework. EEG-related S-TVMs differed between session phases, highlighting the possibility of neurophysiologically measuring relational dynamics. This provided initial evidence that neurophysiological activity can be used to understand differences in the coach-coachee relationship, providing information on the effectiveness of coaching interventions and

facilitating a better transition from education to work.

In [J20] we recorded frontal alpha asymmetry (FAA) and skin conductance level (SCL) from 30 participants who individually took part in a structured job interview to monitor their engagement and stress levels, respectively. Participants were randomly assigned to five different interviews characterised by a pace-full (B, C, D) or stress-full (A, E) style, with each interview consisting of four distinct phases (P1–P4). We found that interviewers B, C and D produced higher engagement levels in candidates. Furthermore, P2 and P4 (related to the explanation of the job and remuneration, respectively) were the most stressful phases, while P1 and P3 (related to the presentation of the company and explanation of the career process, respectively) elicited the greatest engagement.

In [C4] we investigated spectral differences in resting-state EEG associated with individual emotional styles to support the psychometric reliability of the Emotional Style Questionnaire using associated neurophysiological measurements. A sample of 121 healthy participants was enrolled. In the alpha and beta bands, Social Intuition showed significant and large effect sizes in parietal and parieto-occipital regions, as well as a significant and large effect size in the gamma band in the prefrontal region. In the beta band, Attention showed a significant and large effect size in the parieto-occipital region.

In [A28] we proposed a preliminary study investigating the cognitive and emotional effects of specific training on risk perception and its impact on workplace safety. We computed EEG-based neurocognitive indicators of attention, cognitive overload and prefrontal asymmetry; skin conductance and heart-rate variation for physiological analysis; and eye-tracking measures to study visual attention during the presentation of ad hoc video and photographic stimuli designed to assess perceived risk in a sample of 12 trained and 12 untrained participants. A risk-perception questionnaire was also administered. We expected differences between trained and untrained personnel, particularly in terms of risk perception, neurophysiological activation, cognitive overload and attention.

In [C3] we investigated the effects of wearing coloured lenses during video-game play by recording several EEG- and SC-related metrics from a group of 70 participants. Using a within-subjects experimental design, we tested three conditions: no lenses (NoL), neutral lenses (NDL), and yellow lenses (YL). Workload in the YL condition was nearly significantly lower than in the NDL condition, while attention was higher than in the NoL condition for both NDL and YL. We found no differences in arousal or valence. The results suggested that yellow lenses may reduce cognitive load and increase attention.

In [A6] we proposed a preliminary study investigating the cognitive and emotional impact of an AI-based Computer-Assisted Interpreting (CAI) tool during remote simultaneous interpreting. Graduate students in Conference Interpreting performed three five-minute remote interpreting tasks under controlled laboratory conditions: traditional interpreting (control), interpreting with full speech transcription, and interpreting with integrated AI-based CAI support. Neurophysiological data (EEG, skin conductance, heart rate) and psychometric measures (NASA-TLX, SDSSQ, Time Perception) were collected to assess cognitive load, attention, and stress. We expect the use of support tools to increase attentional demands and cognitive load while reducing perceived stress because of their facilitative role. Transcription conditions are expected to produce intermediate effects.

Consumer Neuroscience & Affective Neuroscience

This research line focused on the analysis of neuropsychological data to evaluate emotional responses and preferences elicited by video advertising and product packaging.

In [J18] we assessed eye-tracking metrics (Entry Time and Time Spent) in relation to perceived quality, willingness to pay (WTP), and perceived territoriality for several traditional Sicilian prod-

ucts. Twenty-four images of seafood packages from 12 different producers were presented to 20 students, while subjective evaluations of perceived quality, price and WTP were recorded. Two AOIs were defined within each stimulus, one containing only the brand and the other the product name. Products with clear references to territoriality showed greater Time Spent in the product AOI than products without such references. Perceived quality, price, and WTP showed significant effects on Time Spent in the product AOI, which also positively correlated with ratings of perceived quality, price, and WTP.

In [J21] we assessed the effectiveness of a UNICEF bequests campaign in terms of emotional responses, comparing different creative proposals to optimise communication. The experiment involved 70 participants (35 males) and compared two different spots and flyers. The progeny factor was introduced to assess differences in the impact of bequests depending on the presence or absence of potential heirs. We found differences in both the EEG Approach-Withdrawal Index (AWI), reflecting cognitive involvement, and SC, reflecting emotional activation, particularly among participants without children (the target audience) and for a specific spot linked to the possibility of continuing to live after death.

In [J15] we assessed two video advertisements for certified dairy products to identify the most effective in terms of likeability and the ability to activate mnemonic processes. The commercials differed in their storytelling: the first primarily elicited positive emotions, while the second generated negative emotions during the early stages and positive emotions at the end. Based on previous literature on emotional sequencing in social advertising, we hypothesised that the second commercial would generate a better overall emotional response and activate stronger mnemonic processes. Our results partially supported the research hypotheses, providing useful insights for both marketers and future research.

In [A29] we investigated the role of “mixed emotions” in social campaigns by collecting personality variables and neurophysiological measurements. Our aim was to examine whether personality variables influence the emotions experienced when watching social campaigns and, consequently, willingness to donate. We presented a spot selected during a pilot study to 150 participants. Personality variables and neurophysiological data (EEG) were collected during stimulus presentation. The results, still being processed, will allow us to test two hypotheses. First, resilience affects the experience of mixed emotions while watching the campaign; therefore, higher scores on this dimension are expected to positively affect the propensity to donate. Second, in individuals with low resilience, the EEG withdrawal index detected during exposure to the problem is expected to persist longer than in individuals with high resilience.

In [J16] we discussed Virtual Reality (VR) and Augmented Reality (AR) from a neuromarketing perspective. First, we briefly described VR and AR and their applications in retail, visual merchandising, healthcare marketing and tourism. We focused on how AR can produce immersive and pleasant emotions, increasing the likelihood of users wishing to repeat the experience. We also introduced neuromarketing tools as a means of overcoming some limitations of traditional marketing methodologies by directly measuring consumers’ emotional reactions. We highlighted the lack of neuromarketing studies based on AR frameworks and proposed a joint roadmap for future research in neuromarketing and AR, focusing on how neuromarketing tools can be used to monitor users’ emotional and cognitive states.

In [A25] and [J12] we investigated the role of emotional factors in decision-making elicited by infomercials, focusing on the impact of the seller’s gender and communication style on call-to-action and purchase intent. Forty participants were divided into two groups, each exposed to one of two infomercials. EEG, SC, and PPG signals were recorded during viewing, together with eye-tracking data. After viewing, participants completed a self-report questionnaire. Two communication styles (Lively and Friendly) were identified, and the Affective Trustworthiness and Perceived Trustworthi-

ness scales were used to assess perceived trustworthiness of the seller. The results indicated that seller gender and communication style significantly affected participants’ responses in terms of call-to-action, expected convenience, hospitality and seller performance, as well as the EEG-based WPI (Willingness-to-Pay Index) and AWI (Approach-Withdrawal Index).

In [J10], [A22], [A23], [A27] we compared emotional and cognitive aspects of the shopping experience on metaverse and traditional e-commerce (EC) platforms using EEG, SC, BVP and self-report measures. Twelve participants purchased an identical product once within the Second Life (SL) platform and once through EC. After each task, questionnaires were administered to measure perceived enjoyment, informativeness, ease of use, cognitive effort, and flow. SL showed greater cognitive engagement and memorisation, but also higher workload and a lower emotional response compared with EC, resulting in an overall poorer shopping experience.

In [A17], [A24] we investigated emotional and cognitive reactions during the observation, tasting and enjoyment of a pasta dish. To this end, we recorded neurophysiological measures based on EEG, skin conductance and heart rate, together with self-reports collected using the Self-Assessment Manikin (SAM). Regarding the neurophysiological Memorisation (MI), Cognitive Engagement (BATR) and Emotional (EI) indices, we observed significantly higher values during actual tasting. During the tasting phase, we also found a significant positive correlation between BATR and SAM valence. In [J6] we reanalysed the data using Linear Mixed Models. We observed increased BATR during the sight and tasting phases, as well as a peak in MI during tasting. EI was higher during tasting and savouring than during the initial sight phase. While neuroscientific data indicated no flavour-related differences, self-report measures showed some differences in the emotional responses elicited.

In [A26] we investigated the relationship between consumers’ implicit association with a well-known luxury fashion brand, measured using the Single Category Implicit Association Test (SC-IAT) “self-brand”, and the emotional/cognitive experience elicited by viewing an institutional video, measured using EEG. The results showed a significant correlation between SC-IAT scores and explicit self-report measures used as controls. Furthermore, participants with a stronger implicit and explicit association between self and brand showed greater cognitive engagement, memorisation and emotional activation while viewing the video. However, a negative emotional valence was observed among those with a stronger association between brand and self, suggesting a potential moderating effect of perceived consistency between brand values and video content.

In [C2], [A16] we compared emotional responses to real versus AI-generated pictures and examined the effect of participants’ awareness of image origin using neurophysiological measures (EEG and SC) and self-reports (Self-Assessment Manikin – SAM). Awareness that an image was AI-generated produced an expectation effect in both EEG and SC, significantly affecting motivational approach and arousal. Additionally, AI-generated image origin itself increased arousal, probably because of the hyper-realism, complexity, and novelty of the stimuli.

In [A13]–[A15] we investigated the effects of visual and olfactory nudges, alone and in combination, on university students’ vending-machine snack choices. EEG, skin conductance (SC) and eye-tracking (ET) data were recorded to estimate emotional and cognitive reactions. Eighty-eight university students were exposed to different nudging conditions while observing and selecting snacks from a vending-machine replica. The results showed that visual and olfactory nudges significantly increased the selection of healthier snacks, whereas the combined condition did not. SC data suggested that the nudges may induce relaxation, potentially promoting healthier choices, while the EEG Approach-Withdrawal Index indicated a lower cognitive orientation towards snacks in the combined condition. ET data showed that positioning healthy snacks on a lower shelf distracted attention from them. The combined nudging condition also delayed attention towards healthier products, suggesting its ineffectiveness. In [J5] we reanalysed the data using Linear Mixed Models.

SC suggested that olfactory and combined nudge interventions induced relaxation, potentially promoting healthier choices. EEG showed a reduced motivational approach towards snacks when both nudges were combined. Eye-tracking indicated that individuals paid greater attention to healthy snacks when positioned at eye level, although this effect also depended on the nudge condition.

In [A18] we investigated the doctor-patient relationship in chronic disease contexts using self-reports and neurophysiological measures. Using a preliminary questionnaire ($N = 132$), strengths and weaknesses of the relationship were identified across the dimensions of responsiveness, perceived competence, involvement, trust, empathy, and relationship depth. Subsequently, 20 experimental participants (10 patients and 10 controls) watched eight videos depicting moments of doctor-patient interaction characterised by two different communication styles identified from the questionnaire: one preferable ($n = 4$) and one inadvisable ($n = 4$). During viewing, emotional responses were estimated using skin conductance (SC), heart rate (HR) and prefrontal asymmetry (AWI). The results showed that the advisory communication style was more effective in evoking positive emotions and greater involvement, leading to a general perceived improvement in the relationship with the physician.

In [J7], [A20] we explored differences in hedonic perception and communication impact, mediated by semantic priming, between a traditional cheese product and one made from an agro-industrial by-product. For each cheese product, twenty-two participants performed a simple tasting and a tasting based on the Temporal Dominance of Sensations (TDS) method, both under blind conditions and after priming. During tasting, participants' electrical brain activity was recorded using EEG. The results showed no significant differences in EEG data between the two tasting conditions, suggesting that TDS can be integrated without compromising the quality of bioelectrical measurements. Furthermore, a tendency towards greater product appreciation after communicative priming was observed.

In [A11], [A21] we proposed a preliminary study of cognitive and emotional responses during interaction with three eco-label coffee products administered in randomised order. Seventy-two participants interacted with the products under three experimental conditions: (1) searching for additional information through an Augmented Reality (AR) app, (2) searching for additional information on a web page accessed through QR-code scanning, and (3) not searching for additional information. EEG, skin conductance (SC) and eye-tracking (ET) recordings were collected in each condition. From the results, which are still being processed, we expect sustainability information provided through QR and AR to positively affect perceptions and orientation towards the purchase of environmentally friendly products.

In [A12] we explored neural correlates associated with perspective-taking in response to interactions presented in advertisements and tested their ability to predict commercial effectiveness. Nineteen participants were exposed to a series of commercials while their brain activity was recorded using EEG and subsequently answered questions measuring their degree of perspective-taking and subjective evaluation of the commercial (liking, perceived effectiveness, willingness to pay, and intention to purchase). Results suggest that neural activity in the theta band may be involved in perspective-taking processes related to characters' actions in commercials and highlight the need for further research on this topic.

In [W1], [A4] we explored the potential of Augmented Reality (AR)-based technologies (AR-based and QR-scanning apps) as tools for effectively communicating sustainability information in retail environments. We measured cognitive workload through electroencephalography and emotions through skin conductance and heart rate, together with self-report evaluations of green company orientation (GCO), green product information (GPI), and Brand Familiarity, in a sample of 72 participants in a retail setting. Both AR and QR codes enhanced the effectiveness of sustainability communication compared with traditional product packaging. Both digital technologies evoked more positive emotions and led to higher perceptions of GCO and GPI, while AR applications were also

associated with higher cognitive workload. Additionally, Brand Familiarity emerged as a key factor in increasing GCO and GPI.

In [J1] we explored the EEG correlates of cognitive and affective perspective-taking and the effectiveness of video advertisements (ADVs). We recorded EEG from 19 participants while they viewed food and beverage ADVs featuring product interactions. We computed channel-wise spectral power and measures from both original and Minimum Spanning Tree networks and correlated them with self-report scales. Cognitive perspective-taking negatively correlated with frontal θ power and positively with the α clustering coefficient. Effectiveness positively correlated with the α clustering coefficient and degree distribution.

In [J2], [A3] we explored the effectiveness and impact on decision-making of two types of AR applied to packaging. A within-subjects design involving 20 participants was assessed using neurophysiological (EEG, HR and SC) and self-report measures (Augmented Reality Immersion, Perceived Informativeness and Authenticity). Neurophysiological findings suggested that the immersive AR application elicited greater emotional and partially cognitive engagement, as well as higher perceived immersion according to self-reports.

In [W2] we explored how wine-bottle closures (cork vs. screw cap) can influence both implicit emotional and cognitive responses and explicit evaluations of wine-tasting experiences through multisensory integration. Forty-four participants tasted wines preceded by priming cues (auditory vs. visual) indicating the wine closure type, while EEG, SC and HR signals were collected together with self-reports of wine quality, celebration suitability, willingness to pay, and perceived aromas. Neurophysiological results showed that knowing a wine had a cork closure enhanced memory processes, while presenting this information through auditory cues appeared to trigger stronger emotional reactions. Self-reports indicated that wines primed with a cork closure and label information were rated as higher in quality, more suitable for celebrations, and associated with greater willingness to pay than screw-cap wines primed with auditory cues. Additionally, interactions between closure and priming type influenced perceived wine aromas.

In [A8]–[A10] we compared cognitive and emotional responses to real and virtual cultural heritage environments using a within-subjects design involving 20 participants, with EEG, HR, eye-tracking, and self-report measures. Participants experienced both a physical and a virtual visit to Palazzo Bernabei (Assisi, Italy). The physical visit was associated with higher memorisation (EEG), more positive emotions (HR), and greater satisfaction and loyalty (self-reports). In contrast, the virtual visit elicited increased attention (ET), engagement and cognitive load (EEG). These results provide insights into the design of engaging digital cultural experiences and emphasise the complementary value of physical and virtual strategies.

In [A7] we proposed a preliminary investigation into how the nature of artistic stimuli and communicative framing influence aesthetic and neurophysiological responses in virtual museum environments. Participants from Generations Y and Z experienced two types of virtual museums: (A) digital reproductions of physical artworks and (B) natively digital exhibitions (e.g., crypto-art, NFTs). Participants were randomly assigned to one of three framing conditions preceding the experience: no brief, congruent brief, or incongruent brief. Neurophysiological measures (EEG, eye-tracking, galvanic skin response, and heart rate) were recorded together with post-experience questionnaires assessing aesthetic evaluation, perceived authenticity, immersion, time perception, and behavioural intention. We expect congruence between communicative framing and artistic stimulus to significantly modulate both neurophysiological and subjective responses.

In [C1] we investigated the implicit dimensions of brand reputation using the N400 component of event-related potentials (ERPs), a neural marker of semantic congruence. Ninety-seven participants were exposed to 16 real brands from eight market sectors, each paired with positive and negative reputational attributes, while EEG activity was recorded. A differential Δ N400 index was computed

to assess the semantic integration of brand–attribute associations. Preliminary results showed that this approach could discriminate brands and sectors according to their implicit reputation, although significant effects emerged for only three brands and one sector. The method also allowed us to identify specific reputational vulnerabilities, providing useful managerial insights and suggesting the integration of ERP measures with behavioural and survey-based tools.

Electronic Design

This research line focused on the design of open-source electronic systems for EEG research as effective low-cost alternatives to commercially available devices.

In [J22] we designed, developed and tested an EEG synchronisation box (ESB) able to condition and distribute a synchronisation signal (analogue and digital) to different devices simultaneously. The ESB can be assembled by individuals with limited soldering skills using a protoboard and point-to-point wiring. It represents a cost-effective alternative to commercially available synchronisation boxes (the overall cost was approximately 35 EUR), while preserving similar electrical and functional characteristics (e.g., input-voltage range and minimum input/output pulse duration).

Non-local Communication

This research line focused on the analysis of EEG data from pairs of spatially separated individuals to investigate the possibility of non-local brain-to-brain communication, as well as non-local mind-matter interactions with electronic devices.

In [J26] we analysed the EEG activity of two spatially separated human partners when one member of the pair received stimulation (sender, S) and the second was connected only mentally with the first (receiver, R). Seven selected S-R couples with a long-standing friendship and an ability to maintain focused mental concentration were divided into two groups located in separate laboratories approximately 190 km apart. Using a Support Vector Machine (SVM) classifier, we detected with high accuracy coincidences in the sequence of events of the stimulation protocol between the EEG activity of the S-R pair. Additionally, we found a positive correlation between the EEG power spectra of the twenty pairs.

In [J24] we investigated EEG correlates in twenty-five spatially separated pairs of participants. One member of each pair (sender, S) received a light/audio stimulus, while the other member (receiver, R) sat in an isolated sound-proof room. Analysis of Event-Related Potentials associated with sensory stimulation using traditional averaging methods showed a distinct P300 component, but only in the EEG activity of the S member. By applying the GW6 algorithm [J25], a weak but robust response was also detected in the EEG activity of the R member.

In [J14] we explored the possibility of (a) mentally producing, at a geographical distance, pre-defined images on digital sensors within modern professional-grade photographic equipment, with (b) sufficient resolution to be objectively verified using a Structural Similarity Index and specialised pattern-recognition software. Three participants experienced in distant ‘mind-matter interaction’ techniques completed a total of 49 trials. In 6 out of 49 trials (12.2

Miscellaneous

The following articles cannot be grouped into a single research line, as they address different areas, including quantitative consumer-behaviour analysis and psychometrics.

In [J19] we explored differences between two groups, Italian and French, in the perception of and intention to buy products with certification marks by analysing a CAWI survey. We tested a

Multi-Group Structural Equation Model assessing the mediation of Perceived Product Safety (PPS) between Packaging with reference to Territoriality (PT) and Intention to Buy (IB). We showed that, in both groups, PT was positively associated with IB and PPS, and PPS was positively associated with IB. The main difference was that mediation by PPS was present only in the Italian group.

In [A30] we proposed a preliminary Italian adaptation of the Emotional Style Questionnaire (ESQ) by Kesbir and colleagues. Following a backward/forward translation procedure, we translated the 24 ESQ items into Italian and administered them to a sample of 200 university students. Exploratory factor analysis confirmed a reliable six-factor structure. Confirmatory factor analysis showed good fit indices ($\chi^2(169) = 261.140$, $p < 0.001$, CFI = 0.93, TLI = 0.90, RMSEA = 0.05, SRMR = 0.06), with all items loading significantly on their corresponding factors.

In [J11] we explored the impact of product communication (CP), territory (PT), and sustainability (SP) on perceived product safety (PPS) for traditional food products (TFP) in the dairy sector. A structural equation model was applied to a sample of 1,079 young Italian cheese consumers. The results confirmed the hypothesised impact of CP on PPS through the mediation of PT and, particularly, SP. SP played a crucial communicative role in the model, demonstrating its ability to enhance perceptions of TFP safety. This research contributes to knowledge in the field of TFPs by highlighting communication and sustainable packaging as key factors conveying healthiness, nutritiousness, and perceived safety, consequently supporting wider diffusion of these products in the market.

In [J8], [A5] we explored the factors that should be considered to increase consumers' use of 19 cosmetics apps, based on the well-known Technology Acceptance Model (TAM) applied to AR. An online self-report questionnaire was used to assess Perceived Informativeness (PI) and Perceived Ease of Use (PEOU) as potential predictor variables, Behavioral Intention (BI) as an outcome variable, together with Trust, Doubt, Make-up Involvement (MI), and Perceived Diagnosticity (PD). Perceived Enjoyment (PE) was included as a mediating variable. The hypothesised structural equation model (SEM) indicated that PI was a strong predictor across the model, while PE acted as an important mediator. Results for Trust and Doubt suggested that moderate levels of PE and PEOU were preferable; otherwise, Doubt tended to increase and Trust to decrease. PD and BI were positively affected by the AR experience, and their coexistence appeared important for the optimal and effective use of the application. Finally, the mediating role of PE supported the finding that consumers' BI increases with the information they obtain.

9 Publications

Journal Articles

§: corresponding authorship, †: co-first authorship

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